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Contoh 6 $\rightarrow f(x_1, x_2) = -x_1^2 + x_2^2$

vektor gradien $f(x)$:

$$\nabla f(x) = \begin{pmatrix} \frac{\partial f(x)}{\partial x_1} \\ \frac{\partial f(x)}{\partial x_2} \end{pmatrix} = \begin{pmatrix} -2x_1 \\ 2x_2 \end{pmatrix}$$

Matrix Hessian $f(x)$:

$$H(x) = \begin{bmatrix} \frac{\partial^2 f(x)}{\partial x_1 \partial x_1} & \frac{\partial^2 f(x)}{\partial x_1 \partial x_2} \\ \frac{\partial^2 f(x)}{\partial x_2 \partial x_1} & \frac{\partial^2 f(x)}{\partial x_2 \partial x_2} \end{bmatrix}$$
$$= \begin{pmatrix} -2 & 0 \\ 0 & 2 \end{pmatrix}$$

Det LPM-1 $H(x) = \det(-2) = -2 < 0$ trap x

Det LPM-2 $H(x) = \det \begin{bmatrix} -2 & 0 \\ 0 & 2 \end{bmatrix} = -4 < 0$ trap x

- Tanda det LPM-1 adalah $\ominus$ trap x , dan
tanda det LPM-2 adalah $\oplus$ trap x , maka
 $H(x)$ tidak definit

$\rightarrow f(x)$ adalah fungsi tidak convex tidak covey

Contoh 7 $\rightarrow f(x_1, x_2) = x_1^3 + 2x_2^2 + 3x_1 + 4x_2 + 5$

Vektor gradien $f(x) : \begin{bmatrix} \frac{\partial f(x)}{\partial x_1} \\ \frac{\partial f(x)}{\partial x_2} \end{bmatrix} = \begin{bmatrix} 3x_1^2 + 3 \\ 4x_2 + 4 \end{bmatrix}$

$\nabla f(x) = \begin{bmatrix} \frac{\partial f(x)}{\partial x_1} \\ \frac{\partial f(x)}{\partial x_2} \end{bmatrix}$

Matriks Hessian :

$$H(x) = \begin{bmatrix} \frac{\partial^2}{\partial x_1 \partial x_1} f(x) & \frac{\partial^2}{\partial x_1 \partial x_2} f(x) \\ \frac{\partial^2}{\partial x_2 \partial x_1} f(x) & \frac{\partial^2}{\partial x_2 \partial x_2} f(x) \end{bmatrix}$$

$$= \begin{bmatrix} 6x_1 & 0 \\ 0 & 4 \end{bmatrix}$$

Det LPM-1 $H(x) = \det(6x_1) = 6x_1 \begin{cases} < 0, x_1 < 0 \\ = 0, x_1 = 0 \\ \neq 0, x_1 \neq 0 \end{cases}$

Det LPM-2 $H(x) : \det \begin{bmatrix} 6x_1 & 0 \\ 0 & 4 \end{bmatrix} = 24x_1 \begin{cases} < 0, x_1 < 0 \\ = 0, x_1 = 0 \\ > 0, x_1 > 0 \end{cases}$

$\rightarrow$ Det LPM 1 dan LPM 2 dapat berturut-turut,

$\oplus \ominus$, atau $\ominus$ untuk tiap x , maka $H(x)$ adalah ~~definit~~ tak definit

$\rightarrow f(x)$ adalah fgs: tak cembung tak cekung